

Week/Lecture 11

Linear First Order ODEsReview

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$I.F = e^{\int P(x) dx}$$

Example

$$\frac{dy}{dx} + y = -\frac{1}{2}x \quad \leadsto (1)$$

$$P(x) = 1 \quad Q(x) = -\frac{1}{2}x$$

$$I.F = e^{\int P(x) dx} = e^{\int 1 dx} = e^x \quad \leadsto (2)$$

use (2) in (1)

$$e^x \frac{dy}{dx} + e^x y = -\frac{1}{2}x e^x$$

$$\boxed{\frac{d}{dx}(e^x y) = -\frac{1}{2}x e^x}$$

Integrate w.r.t x

$$\int \frac{d}{dx}(e^x y) dx = -\frac{1}{2} \int x e^x dx$$

$$e^x y = -\frac{1}{2} [x e^x - \int e^x dx] + C$$

$$= -\frac{1}{2} [x e^x - e^x] + C$$

$$e^x y = -\frac{1}{2} (x-1) e^x + C$$

$$y = \frac{-\frac{1}{2} (x-1) e^x}{e^x} + \frac{C}{e^x}$$

$$y = -\frac{1}{2} (x-1) + C e^{-x}$$

Ans.

SECTION 15.2 First-Order Linear Differential Equations

First-Order Linear Differential Equations • Bernoulli Equations • Applications

First-Order Linear Differential Equations

In this section, you will see how integrating factors help to solve a very important class of first-order differential equations—first-order *linear* differential equations.

Definition of First-Order Linear Differential Equation

A first-order linear differential equation is an equation of the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where P and Q are continuous functions of x . This first-order linear differential equation is said to be in **standard form**.

To solve a first-order linear differential equation, you can use an integrating factor $u(x)$, which converts the left side into the derivative of the product $u(x)y$. That is, you need a factor $u(x)$ such that

$$\begin{aligned} u(x)\frac{dy}{dx} + u(x)P(x)y &= \frac{d[u(x)y]}{dx} \\ u(x)y' + u(x)P(x)y &= u(x)y' + yu'(x) \\ u(x)P(x)y &= yu'(x) \\ P(x) &= \frac{u'(x)}{u(x)} \\ \ln|u(x)| &= \int P(x) dx + C_1 \\ u(x) &= Ce^{\int P(x) dx}. \end{aligned}$$

Because you don't need the most general integrating factor, let $C = 1$. Multiplying the original equation $y' + P(x)y = Q(x)$ by $u(x) = e^{\int P(x) dx}$ produces

$$\begin{aligned} y'e^{\int P(x) dx} + yP(x)e^{\int P(x) dx} &= Q(x)e^{\int P(x) dx} \\ \frac{d}{dx} \left[ye^{\int P(x) dx} \right] &= Q(x)e^{\int P(x) dx}. \end{aligned}$$

The general solution is given by

$$ye^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx + C.$$

ANNA JOHNSON PELL WHEELER (1883–1966)

Anna Johnson Pell Wheeler was awarded a master's degree from the University of Iowa for her thesis *The Extension of Galois Theory to Linear Differential Equations* in 1904. Influenced by David Hilbert, she worked on integral equations while studying infinite linear spaces.

THEOREM 15.3 Solution of a First-Order Linear Differential Equation

An integrating factor for the first-order linear differential equation

$$y' + P(x)y = Q(x)$$

is $u(x) = e^{\int P(x) dx}$. The solution of the differential equation is

$$ye^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx + C.$$

STUDY TIP Rather than memorizing this formula, just remember that multiplication by the integrating factor $e^{\int P(x) dx}$ converts the left side of the differential equation into the derivative of the product $ye^{\int P(x) dx}$.

EXAMPLE 1 Solving a First-Order Linear Differential Equation

Find the general solution of

$$xy' - 2y = x^2.$$

Solution The *standard form* of the given equation is

$$y' + P(x)y = Q(x)$$

$$y' - \left(\frac{2}{x}\right)y = x.$$

Standard form

Thus, $P(x) = -2/x$, and you have

$$\int P(x) dx = -\int \frac{2}{x} dx = -\ln x^2$$

$$e^{\int P(x) dx} = e^{-\ln x^2} = \frac{1}{x^2}.$$

Integrating factor

Therefore, multiplying both sides of the standard form by $1/x^2$ yields

$$\frac{y'}{x^2} - \frac{2y}{x^3} = \frac{1}{x}$$

$$\frac{d}{dx} \left[\frac{y}{x^2} \right] = \frac{1}{x}$$

$$\frac{y}{x^2} = \int \frac{1}{x} dx$$

$$\frac{y}{x^2} = \ln |x| + C$$

$$y = x^2(\ln |x| + C).$$

General solution

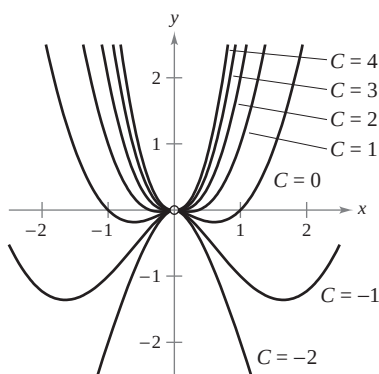


Figure 15.5

Several solution curves (for $C = -2, -1, 0, 1, 2, 3$, and 4) are shown in Figure 15.5.

Now, we discuss some important rules which are often helpful in finding integrating factors of certain equations of special types. These rules are illustrated below (without proofs).

Rule 1. When the non-exact differential equation, $M(x, y) dx + N(x, y) dy = 0$ is homogeneous, and $(Mx + Ny) \neq 0$, then, $\frac{1}{Mx + Ny}$ is an integrating factor of the given differential equation.

Example 3.8.1 Solve the differential equation,

$$(x^2 y - 2xy^2) dx - (x^3 - 3x^2 y) dy = 0 \quad (3.8.1)$$

Solution.

This is a homogeneous differential equation of the form $M dx - N dy = 0$. It is not exact as $\partial M / \partial y \neq \partial N / \partial x$. Also,

$$Mx + Ny = x^3 y - 2x^2 y^2 - x^3 y + 3x^2 y^2 = x^2 y^2 \neq 0.$$

Therefore, according to Rule 1, $1/(Mx + Ny) = 1/x^2 y^2$ is an integrating factor of Eqn (3.8.1). Therefore, on multiplying Eqn. (3.8.1) by $1/x^2 y^2$, we obtain,

$$\left(\frac{1}{y} - \frac{2}{x}\right) dx - \left(\frac{x}{y^2} - \frac{3}{y}\right) dy = 0.$$

This is an exact differential equation. Integrating it by the method given in sec 3.6 we get,

$$\frac{x}{y} - 2 \ln |x| + 3 \ln |y| = c,$$

which is the required general solution of the given differential equation.

Example 3.8.1 a. Solve the differential equation,

$$(x^2 + y^2) dx + 3xy dy = 0. \quad (3.8.1a)$$

Solution.

This is a homogeneous differential equation of the form $M dx + N dy = 0$. It is not exact as $\partial M / \partial y \neq \partial N / \partial x$. Also, $Mx + Ny = x^3 + 4xy^2 \neq 0$. Therefore, according to Rule 1, $1/(Mx + Ny) = 1/(x^3 + 4xy^2)$ is an integrating factor (I.F.) of Eqn. (3.8.1a). On multiplying Eqn (3.8.1a) by this I. F., we get,

$$\frac{x^2 + y^2}{x^3 + 4xy^2} dx + \frac{3xy}{x^3 + 4xy^2} dy = 0. \quad (3.8.1b)$$

This is an exact differential equation. On integrating the second member of this equation (of course w.r.t. y), we obtain,

$$\frac{3}{8} \ln |x^3 + 4xy^2|.$$

Next, we seek a function $f(x)$, of x alone, such that

$$d \left[f(x) + \frac{3}{8} \ln |x^3 + 4xy^2| \right]$$

is the left hand member of equation (3. 8. 1b). We see at once that,

$$f'(x) = -\frac{1}{8x}.$$

Hence, $f(x)$ may be taken as $-\frac{1}{8} \ln |x|$. Therefore, we obtain the solution of the given differential equations as ,

$$-\frac{1}{8} \ln |x| + \frac{3}{8} \ln |x^3 + 4xy^2| = c_1;$$

$$\text{Or, } \frac{1}{8} \left[\ln \left| \frac{1}{x} \right| + \ln |x^3 + 4xy^2|^3 \right] = c_1;$$

$$\text{Or, } \ln |x^3 (x^2 + 4y^2)^3 / x| = 8c_1 = c_2 = \ln e^{c_2};$$

$$\text{Or, } x^2 (x^2 + 4y^2)^3 = c.$$

Note. This problem can also be solved by technique # 2 for homogeneous differential equations.

Rule 2. If the differential equation, $M(x, y) dx + N(x, y) dy = 0$ is of the form,

$$y \cdot f_1(xy) dx + x \cdot f_2(xy) dy = 0, \text{ and, } (Mx - Ny) \neq 0,$$

then the equation has, $\frac{1}{Mx - Ny}$ as an integrating factor.

(Note: f_1 and f_2 are functions of the product xy)

Examples 3. 8. 2. Solve the differential equation,

$$y(xy + 2x^2y^2) dx + x(xy - x^2y^2) dy = 0 \quad (3. 8. 2)$$

Solution.

This differential equation is of the form,

$$y f_1(xy) dx + x f_2(xy) dy = 0,$$

and,

$$(Mx - Ny) = x^2 y^2 + 2x^3 y^3 - x^2 y^2 + x^3 y^3 = 3x^3 y^3 \neq 0.$$

Therefore, by Rule 2, $1/(3x^3 y^3)$ is an integrating factor (I.F.) of Eqn.(3.8.2), which on being multiplied by this I. F. becomes,

$$\left(\frac{1}{x^2 y} + \frac{2}{x}\right)dx + \left(\frac{1}{xy^2} - \frac{1}{y}\right)dy = 0.$$

This is an exact differential equation. On simplification it becomes,

$$\frac{y dx + x dy}{x^2 y^2} + \frac{2}{x} dx - \frac{1}{y} dy = 0;$$

or, $(xy)^{-2} d(xy) + (2/x) dx - (1/y) dy = 0.$

On integration it gives the required solution as,

$$-1/xy + 2 \ln |x| - \ln |y| = c.$$

Rule 3. When the equation, $M(x, y) dx + N(x, y) dy = 0$ is not exact, and

$$\left[\frac{\partial M / \partial y - \partial N / \partial x}{N} \right] \text{ is a function of } x \text{ alone, i.e.,}$$

$$\frac{\partial M / \partial y - \partial N / \partial x}{N} = f(x),$$

then $e^{\int f(x) dx}$ is an integrating factor of the given differential equation.

Examples 3. 8. 3. Solve the differential equation,

$$y dx - x dy = 0. \quad (3. 8. 3)$$

Solution. This equation is of the form, $M dx + N dy = 0$. Here $M = y$, and $N = -x$. Therefore,

$$\frac{\partial M / \partial y - \partial N / \partial x}{N} = \frac{1+1}{-x} = \frac{-2}{x}.$$

Thus,

$$\text{I. F.} = e^{-\int (2/x) dx} = e^{\ln(1/x^2)} = 1/x^2.$$

On multiplying this I. F. with Eqn.(3.8.3) we get,

$$\frac{y dx - x dy}{x^2} = 0;$$

Or,

$$-d(y/x) = 0.$$

On integration it gives,

$$y/x = c,$$

which is the required solution of the given differential equation.

Rule 4. When the differential equation, $M(x, y) dx + N(x, y) dy = 0$ is not exact,

and $\frac{\partial N / \partial x - \partial M / \partial y}{M}$ is a function of y only, i.e., $\frac{\partial N / \partial x - \partial M / \partial y}{M} = f(y)$,

then, $e^{\int f(y) dy}$ is an integrating factor of the given differential equation.

Example 3. 8. 4. Solve the differential equation,

$$dx + (x/y - \sin y) dy = 0 \quad (3. 8. 4).$$

Solution. Here,

$$\frac{\partial N / \partial x - \partial M / \partial y}{M} = \frac{1/y - 0}{1} = \frac{1}{y}.$$

Therefore, $e^{\int (1/y) dy} = e^{\ln y} = y$ is an I. F of Eqn (3. 8. 4). On multiplying this equation by the I. F we get,

$$y dx + (x - y \sin y) dy = 0,$$

which is an exact differential equation. On integrating it (by the method for exact differential equations) we obtain the solution,

$$yx - (-y \cos y + \int \cos y dy) = c_1;$$

$$\text{or, } xy + y \cos y - \sin y = c.$$

Rule 5. When the differential equation,

$$x^a y^b (y \cdot A dx + x \cdot B dy) = 0, \text{ (a, b, A, B being constants),}$$

is not exact, then $(x^{kA-1-a} y^{kB-1-b})$ is an integrating factor of this differential equation, where k is a constant whose value is to be determined.

Examples 3. 8. 5. Solve the differential equation,

$$(y^3 - 2x^2y) dx + (2xy^2 - x^3) dy = 0. \quad (3. 8. 5)$$

Solution. On rearranging the terms, Eqn. (3.8.5) becomes,

$$y^2 (y \, dx + 2x \, dy) - x^2 (2y \, dx + x \, dy) = 0, \quad (3.8.6)$$

i.e., $x^0 y^2 (y \cdot 1 \, dx + x \cdot 2 \, dy) - x^2 y^0 (y \cdot 2 \, dx + x \cdot 1 \, dy) = 0.$

Both the terms of this equation are of the form, $x^a y^b (y \cdot A \, dx + x \cdot B \, dy)$. For the first term, $a = 0, b = 2, A = 1, B = 2$. Thus $x^{k-1} y^{2k-3}$ is its I.F. For the second term, $a = 2, b = 0, A = 2, B = 1$, so that $x^{2k'-3} y^{k'-1}$ is its I. F. Both of these *integrating factors* are same if,

$$k - 1 = 2k' - 3, \text{ and } 2k - 3 = k' - 1.$$

Solving these equations simultaneously for k and k' , we obtain : $k = 2 = k'$. Therefore, xy is the common I.F. for both the terms of Eqn.(3.8.6). Accordingly the given differential equation becomes exact on being multiplied by this I.F.,

$$(xy^4 - 2x^3y^2) \, dx + (2x^2y^3 - x^4y) \, dy = 0.$$

Here, we see that $\frac{\partial M}{\partial y} = 4xy^3 - 4x^3y = \frac{\partial N}{\partial x}$. Therefore, the above equation is an exact differential equation. Thus by technique 4 we get the solution of this differential equation as,

$$x^2y^4/2 - x^4y^2/2 = c_1;$$

or, $x^2y^2(y^2 - x^2) = c.$

The rules for finding an *integrating factor* of a non-exact differential equation are summarized in the following table:

Table for Rules to Evaluate Integrating Factors

<u>Rule#</u>	<u>Differential Eqn.</u>	<u>Form of D.E.</u>	<u>Condition</u>	<u>I.F.</u>
1	$M \, dx + N \, dy = 0$	Homogeneous	$Mx + Ny \neq 0$	$\frac{1}{Mx + Ny}$
2	$M \, dx + N \, dy = 0$	$y f_1(xy) \, dx + x f_2(xy) \, dy = 0$	$Mx - Ny \neq 0$	$\frac{1}{Mx - Ny}$
3	$M \, dx + N \, dy = 0$	-	$\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} = f(x)$	$e^{\int f(x) \, dx}$
4	$M \, dx + N \, dy = 0$	-	$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} = f(y)$	$e^{\int f(y) \, dy}$
5	$x^a y^b (yA \, dx + xB \, dy) = 0$ (a, b, A, B being constants)	-	-	$x^{kA-1-a} y^{kB-1-b}$ (k to be determined)

EXERCISE - 3.5

(SOLUTION WITH THE HELP OF INTEGRATING FACTOR)

Show that the given function is an integrating factor of the following differential equation, and solve:

1. $2y \, dx + x \, dy = 0$, (I. F. = x); $[x^2 y = c]$
2. $x \, dy - y \, dx = 0$, (I. F. = $1/x^2$); $[y = cx]$
3. $\sin y \, dx + \cos y \, dy = 0$, (I. F. = e^x); $[e^x \sin y = c]$

$$M_y = \sin y$$

$$N_x = 0$$

$$\frac{M_y - N_x}{N} = \frac{\sin y}{\cos y} = \tan y$$

$$\int \tan y \, dy = -\ln |\cos y|$$

Find an integrating factor for each of the following differential equations and solve them,

4. $2y \, dx + x \, dy = 0$; $[y = c/x^2, \text{ I. F.} = 1/3xy]$
5. $y - x y' = x + y y'$; $[\tan^{-1} \frac{x}{y} - \ln \sqrt{x^2 + y^2} = c; \text{ I. F.} = \frac{1}{x^2 + y^2}]$
6. $x \, dy - y \, dx = x^2 \, dy$; $[y = xy + cx; \text{ I. F.} = 1/x^2]$
7. $x \, dy - y \, dx = (x^2 + y^2) \, dx$; $[y = x \tan(x + c)]$
8. $y \, dx + x \, dy = xy \, dx$; $[xy = ce^x]$
9. $x \, dy - y \, dx = x^3 \, dx$; $[2y = x^3 + cx]$
10. $dy + \frac{y - \sin x}{x} \, dx = 0$; $[xy + \cos x = c; \text{ I. F.} = 1/x]$
11. $(3xy + y^2) \, dx + (x^2 + xy) \, dy = 0$; $[x^3 y + x^2 y^2/2 = c; \text{ I. F.} = x]$
12. $(x^2 + y^2 + 2x) \, dx + 2y \, dy = 0$; $[e^x (x^2 + y^2) = c; \text{ I. F.} = e^x]$
13. $(x^2 + y^2) \, dx - 2xy \, dy = 0$; $[x^2 - y^2 = cx; \text{ I. F.} = -2/x]$
14. $y' = e^{2x} + y - 1$; $[y = ce^x + 1 + e^{2x}; \text{ I. F.} = e^{-x}]$
15. $(3x^2 y + 2xy + y^3) \, dx + (x^2 + y^2) \, dy = 0$; $[(3x^3 y + 3xy^3 + 3x^2 y + y^3) e^{3x} = c; \text{ I. F.} = e^{3x}]$
16. $(x y^2 + y) \, dx - x \, dy = 0$; $[x/y + x^2/2 = c; \text{ I. F.} = 1/y^2]$
17. $(x^2 y - 2xy^2) \, dx - (x^3 - 2x^2 y) \, dy = 0$; $[x/y + \ln |y/x^2| = c; \text{ I. F.} = y^4]$
18. $(3x^2 y^4 + 2xy) \, dx + (2x^3 y^3 - x^2) \, dy = 0$; $[x^3 y^3 + x^2 = cy; \text{ I. F.} = 1/y^2]$
19. $(y^4 + 2y) \, dx + (xy^3 + 2y^4 - 4x) \, dy = 0$; $[xy + y^2 + 2x/y^2 = c; \text{ I. F.} = 1/y^3]$
20. $y \cos x \, dx + 2 \sin x \, dy = 0$; $[y^2 \sin x = c; \text{ I. F.} = y]$
21. $e^x \, dx + (e^x \cot y + 2y \operatorname{cosec} y) \, dy = 0$; $[e^x \sin y + y^2 = c; \text{ I. F.} = \sin y]$
22. $y \, dx + (2xy - e^{-2y}) \, dy = 0$; $[xe^{2y} - \ln |y| = c; \text{ I. F.} = (e^{2y})/y]$
23. $y(2xy + e^x) \, dx - e^x \, dy = 0$; $[x^2 + e^{xy} = c; \text{ I. F.} = e^{-2y}]$
24. $y \, dx + x \, dy = x^2 y^3 \, dy$; $[xy^3 + cxy + 2 = 0; \text{ I. F.} = 1/x^2 y^2]$

25. $(3y + 4xy) dx - 2x dy = 0;$

$[x^3 e^{4x} = cy^2, I.F = 1/xy]$

26. $(3y + 4xy^2) dx + (2x + 3x^2y) dy = 0;$

$[x^3 y^2 + x^4 y^3 = c, I.F = x^2 y]$

27. $y(xy + 2x^2 y^2) dx + x(xy - x^2 y^2) dy = 0;$

$[\ln|x^2/y| = 1/xy + c; I.F = \frac{1}{x^3 y^3}]$

28. $(3x + 6/y) dx + (x^2/y + 3y/x) dy = 0;$

$[x^3 y + 3x^2 + y^3 = c; I.F = xy]$

Find the integrating factors and solve the following initial value problems:

29. $2 \sin y dx + \cos y dy = 0; y(0) = \pi/2;$

$[e^{2x} \sin y = 1, I.F = \operatorname{cosec} y]$

30. $2 dx + \sec x \cos y dy = 0; y(0) = 0;$

$[2 \sin x + \sin y = 0, I.F = \cos x]$

Bernoulli Differential Equations

In this section we are going to take a look at differential equations in the form,

$$y' + p(x)y = q(x)y^n$$

where $p(x)$ and $q(x)$ are continuous functions on the interval we're working on and n is a real number. Differential equations in this form are called **Bernoulli Equations**.

First notice that if $n = 0$ or $n = 1$ then the equation is linear and we already know how to solve it in these cases. Therefore, in this section we're going to be looking at solutions for values of n other than these two.

In order to solve these we'll first divide the differential equation by y^n to get,

$$y^{-n} y' + p(x)y^{1-n} = q(x)$$

We are now going to use the substitution $v = y^{1-n}$ to convert this into a differential equation in terms of v . As we'll see this will lead to a differential equation that we can solve.

We are going to have to be careful with this however when it comes to dealing with the derivative, y' . We need to determine just what y' is in terms of our substitution. This is easier to do than it might at first look to be. All that we need to do is differentiate both sides of our substitution with respect to x . Remember that both v and y are functions of x and so we'll need to use the chain rule on the right side. If you remember your Calculus I you'll recall this is just [implicit differentiation](#). So, taking the derivative gives us,

$$v' = (1-n)y^{-n}y'$$

Now, plugging this as well as our substitution into the differential equation gives,

$$\frac{1}{1-n}v' + p(x)v = q(x)$$

This is a [linear differential equation](#) that we can solve for v and once we have this in hand we can also get the solution to the original differential equation by plugging v back into our substitution and solving for y .

Let's take a look at an example.

Example 1 Solve the following IVP and find the interval of validity for the solution.

$$y' + \frac{4}{x}y = x^3 y^2 \quad y(2) = -1, \quad x > 0$$

Solution

So, the first thing that we need to do is get this into the "proper" form and that means dividing everything by y^2 . Doing this gives,

$$y^{-2} y' + \frac{4}{x} y^{-1} = x^3$$

The substitution and derivative that we'll need here is,

$$v = y^{-1} \quad v' = -y^{-2} y'$$

With this substitution the differential equation becomes,

$$-v' + \frac{4}{x}v = x^3$$

So, as noted above this is a linear differential equation that we know how to solve. We'll do the details on this one and then for the rest of the examples in this section we'll leave the details for you to fill in. If you need a refresher on solving linear differential equations then go back to that [section](#) for a quick review.

Here's the solution to this differential equation.

$$\begin{aligned} v' - \frac{4}{x}v &= -x^3 & \Rightarrow & \mu(x) = e^{\int -\frac{4}{x}dx} = e^{-4\ln|x|} = x^{-4} \\ \int (x^{-4}v)' dx &= \int -x^{-1} dx \\ x^{-4}v &= -\ln|x| + c & \Rightarrow & v(x) = cx^4 - x^4 \ln x \end{aligned}$$

Note that we dropped the absolute value bars on the x in the logarithm because of the assumption that $x > 0$.

Now we need to determine the constant of integration. This can be done in one of two ways. We can convert the solution above into a solution in terms of y and then use the original initial condition or we can convert the initial condition to an initial condition in terms of v and use that. Because we'll need to convert the solution to y 's eventually anyway and it won't add that much work in we'll do it that way.

So, to get the solution in terms of y all we need to do is plug the substitution back in. Doing this gives,

$$y^{-1} = x^4(c - \ln x)$$

At this point we can solve for y and then apply the initial condition or apply the initial condition and then solve for y . We'll generally do this with the later approach so let's apply the initial condition to get,

$$(-1)^{-1} = c2^4 - 2^4 \ln 2 \quad \Rightarrow \quad c = \ln 2 - \frac{1}{16}$$

Plugging in for c and solving for y gives,

$$y(x) = \frac{1}{x^4(\ln 2 - \frac{1}{16} - \ln x)} = \frac{-16}{x^4(1 + 16 \ln x - 16 \ln 2)} = \frac{-16}{x^4(1 + 16 \ln \frac{x}{2})}$$

Note that we did a little simplification in the solution. This will help with finding the interval of validity.

Before finding the interval of validity however, we mentioned above that we could convert the original initial condition into an initial condition for v . Let's briefly talk about how to do that. To do that all we need to do is plug $x = 2$ into the substitution and then use the original initial condition. Doing this gives,

$$v(2) = y^{-1}(2) = (-1)^{-1} = -1$$

So, in this case we got the same value for v that we had for y . Don't expect that to happen in general if you chose to do the problems in this manner.

Okay, let's now find the interval of validity for the solution. First we already know that $x > 0$ and that means we'll avoid the problems of having logarithms of negative numbers and division by zero at $x = 0$. So, all that we need to worry about then is division by zero in the second term and this will happen where,

$$\begin{aligned} 1 + 16 \ln \frac{x}{2} &= 0 \\ \ln \frac{x}{2} &= -\frac{1}{16} \\ \frac{x}{2} &= e^{-\frac{1}{16}} & \Rightarrow & x = 2e^{-\frac{1}{16}} \approx 1.8788 \end{aligned}$$

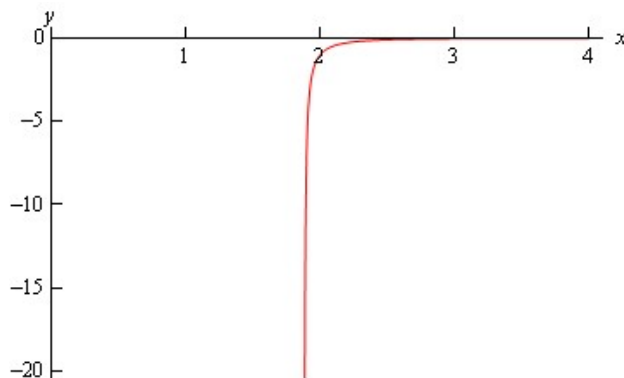
The two possible intervals of validity are then,

$$0 < x < 2e^{-\frac{1}{16}}$$

$$2e^{-\frac{1}{16}} < x < \infty$$

and since the second one contains the initial condition we know that the interval of validity is then $2e^{-\frac{1}{16}} < x < \infty$.

Here is a graph of the solution.



Let's do a couple more examples and as noted above we're going to leave it to you to solve the linear differential equation when we get to that stage.

Example 2 Solve the following IVP and find the interval of validity for the solution.

$$y' = 5y + e^{-2x}y^{-2} \quad y(0) = 2$$

Solution

The first thing we'll need to do here is multiply through by y^2 and we'll also do a little rearranging to get things into the form we'll need for the linear differential equation. This gives,

$$y^2 y' - 5y^3 = e^{-2x}$$

The substitution here and its derivative is,

$$v = y^3 \quad v' = 3y^2 y'$$

Plugging the substitution into the differential equation gives,

$$\frac{1}{3}v' - 5v = e^{-2x} \quad \Rightarrow \quad v' - 15v = 3e^{-2x} \quad \mu(x) = e^{-15x}$$

We rearranged a little and gave the integrating factor for the linear differential equation solution. Upon solving we get,

$$v(x) = ce^{15x} - \frac{3}{17}e^{-2x}$$

Now go back to y 's.

$$y^3 = ce^{15x} - \frac{3}{17}e^{-2x}$$

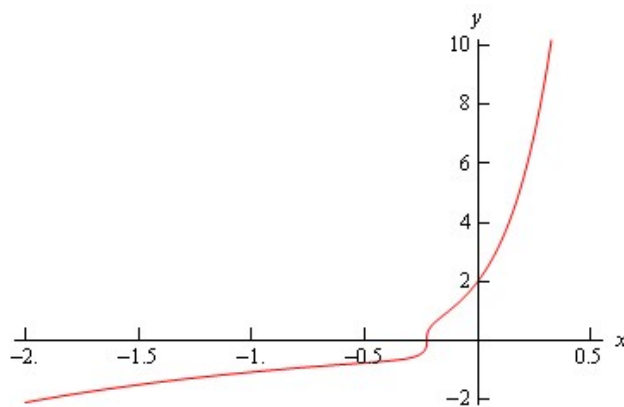
Applying the initial condition and solving for c gives,

$$8 = c - \frac{3}{17} \quad \Rightarrow \quad c = \frac{139}{17}$$

Plugging in c and solving for y gives,

$$y(x) = \left(\frac{139e^{15x} - 3e^{-2x}}{17} \right)^{\frac{1}{3}}$$

There are no problem values of x for this solution and so the interval of validity is all real numbers. Here's a graph of the solution.



Example 3 Solve the following IVP and find the interval of validity for the solution.

$$6y' - 2y = xy^4 \quad y(0) = -2$$

Solution

First get the differential equation in the proper form and then write down the substitution.

$$6y^{-4}y' - 2y^{-3} = x \quad \Rightarrow \quad v = y^{-3} \quad v' = -3y^{-4}y'$$

Plugging the substitution into the differential equation gives,

$$-2v' - 2v = x \quad \Rightarrow \quad v' + v = -\frac{1}{2}x \quad \mu(x) = e^x$$

Again, we've rearranged a little and given the integrating factor needed to solve the linear differential equation.

Upon solving the linear differential equation we have,

$$v(x) = -\frac{1}{2}(x-1) + ce^{-x}$$

Now back substitute to get back into y 's.

$$y^{-3} = -\frac{1}{2}(x-1) + ce^{-x}$$

Now we need to apply the initial condition and solve for c .

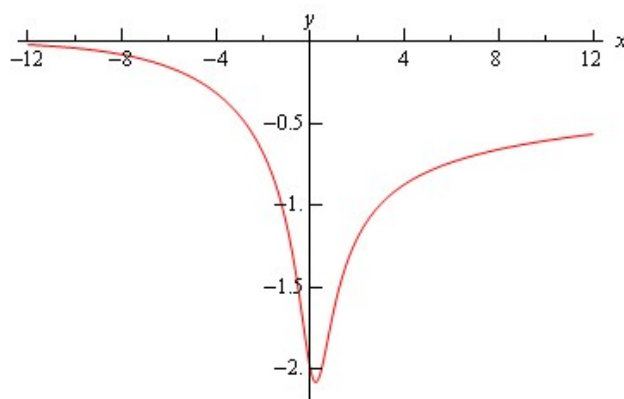
$$-\frac{1}{8} = \frac{1}{2} + c \quad \Rightarrow \quad c = -\frac{5}{8}$$

Plugging in c and solving for y gives,

$$y(x) = -\frac{2}{(4x - 4 + 5e^{-x})^{\frac{1}{3}}}$$

Next, we need to think about the interval of validity. In this case all we need to worry about it is division by zero issues and using some form of computational aid (such as Maple or Mathematica) we will see that the denominator of our solution is never zero and so this solution will be valid for all real numbers.

Here is a graph of the solution.



To this point we've only worked examples in which n was an integer (positive and negative) and so we should work a quick example where n is not an integer.

Example 4 Solve the following IVP and find the interval of validity for the solution.

$$y' + \frac{y}{x} - \sqrt{y} = 0 \qquad y(1) = 0$$

Solution

Let's first get the differential equation into proper form.

$$y' + \frac{1}{x}y = y^{\frac{1}{2}} \qquad \Rightarrow \qquad y^{-\frac{1}{2}}y' + \frac{1}{x}y^{\frac{1}{2}} = 1$$

The substitution is then,

$$v = y^{\frac{1}{2}} \qquad v' = \frac{1}{2}y^{-\frac{1}{2}}y'$$

Now plug the substitution into the differential equation to get,

$$2v' + \frac{1}{x}v = 1 \qquad \Rightarrow \qquad v' + \frac{1}{2x}v = \frac{1}{2} \qquad \mu(x) = x^{\frac{1}{2}}$$

As we've done with the previous examples we've done some rearranging and given the integrating factor needed for solving the linear differential equation. Solving this gives us,

$$v(x) = \frac{1}{3}x + cx^{-\frac{1}{2}}$$

In terms of y this is,

$$y^{\frac{1}{2}} = \frac{1}{3}x + cx^{-\frac{1}{2}}$$

Applying the initial condition and solving for c gives,

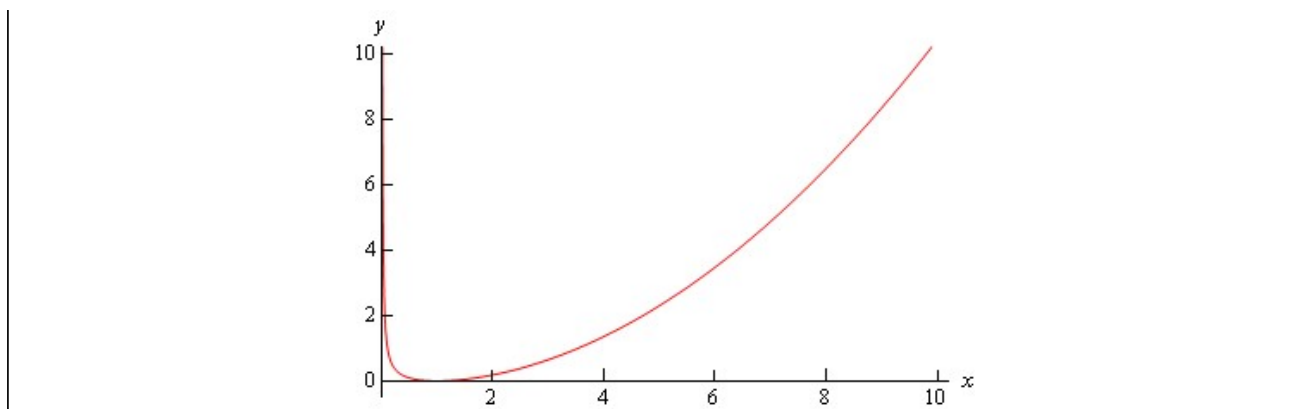
$$0 = \frac{1}{3} + c \qquad \Rightarrow \qquad c = -\frac{1}{3}$$

Plugging in for c and solving for y gives us the solution.

$$y(x) = \left(\frac{1}{3}x - \frac{1}{3}x^{-\frac{1}{2}} \right)^2 = \frac{x^3 - 2x^{\frac{3}{2}} + 1}{9x}$$

Note that we multiplied everything out and converted all the negative exponents to positive exponents to make the interval of validity clear here. Because of the root (in the second term in the numerator) and the x in the denominator we can see that we need to require $x > 0$ in order for the solution to exist and it will exist for all positive x 's and so this is also the interval of validity.

Here is the graph of the solution.



Differential Equations (Notes) / First Order DE's / Bernoulli Differential Equations (Notes) [\[Notes\]](#)

[\[Contact Me\]](#) [\[Downloads\]](#) [\[Privacy Statement\]](#) [\[Site Help\]](#) [\[Site Map\]](#) [\[Terms of Use\]](#) [\[Switch to Mobile View\]](#)

© 2003 - 2018 Paul Dawkins